

IDENTIFYING INFORMATION:

NAME: Borrelli, R. A.

ORCID iD: <https://orcid.org/0000-0003-0274-9253>

POSITION TITLE: Associate Professor

PRIMARY ORGANIZATION AND LOCATION: University of Idaho - Idaho Falls Center for Higher Education, Department of Nuclear Engineering and Industrial Management, Idaho Falls, Idaho, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of California-Berkeley, Berkeley, California, United States	PHD	08/1999 - 12/2006	Nuclear Engineering
Worcester Polytechnic Institute, Worcester, Massachusetts, United States	MS	08/1996 - 05/1999	Civil/Environmental Engineering
Worcester Polytechnic Institute, Worcester, Massachusetts, United States	BS	08/1992 - 05/1996	Mechanical/Nuclear Engineering

Appointments and Positions

2021 - present Associate Professor, University of Idaho - Idaho Falls Center for Higher Education, Department of Nuclear Engineering and Industrial Management, Idaho Falls, Idaho, United States

2025 - present Affiliate Faculty – Electrical & Computer Engineering, University of Idaho, Idaho Falls, Idaho, United States

2019 - present Professional Engineer Faculty Restricted, State of Idaho, Idaho Falls, Idaho, United States

2019 - present Coordinator, Nuclear Power Plant Decommissioning and Used Fuel Management Professional Certificate, University of Idaho, Idaho, United States

2019 - present Affiliate, Boise State University Energy Policy Center, Boise, Idaho, United States

2015 - 2021 Assistant Professor, University of Idaho - Idaho Falls Center for Higher Education, Department of Nuclear Engineering and Industrial Management, Idaho Falls, Idaho, United States

2012 - 2015 Adjunct Professor, Diablo Valley Community College, Department of Architecture and Engineering, Pleasant Hill, California, United States

2009 - 2012 Postdoctorate Researcher, University of California-Berkeley, Department of Nuclear Engineering, Berkeley, California, United States

2007 - 2009 Research Associate, The University of Tokyo, Department of Nuclear Engineering/Management, Tokyo, Not Applicable, N/A, Japan

Products

Products Most Closely Related to the Proposed Project

1. Peterson J, Haney M, Borrelli RA. An overview of methodologies for cyber security vulnerability assessments conducted in nuclear power plants. Nuclear Engineering and Design. 2019; 346:75.
2. Lee J, Shigrekar A, Borrelli R. Hazard and operability analysis of a pyroprocessing facility. Nuclear Engineering and Design. 2019; 348:131.
3. Redfoot EK., Verner KM., Borrelli RA.. Applying analytic hierarchy process to industrial process design in a Nuclear Renewable Hybrid Energy System. Progress in Nuclear Energy. 2022 January; 145:104083.
4. Mena P, Borrelli RA., Kerby L. Expanded Analysis of Machine Learning Models for Nuclear Transient Identification Using TPOT. Nuclear Engineering and Design. 2022; 390:111694.
5. Root SJ, Zhao H, Borrelli RA, McKellar MG. Thermodynamic Analysis on Xenon Stripping to Shorten Restart Time in Molten Salt Microreactors. Nuclear Engineering and Design. 2023; 414:112606.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Carter J, Borrelli RA. Neutron physics study of an integral molten salt reactor using Monte Carlo N-Particle code. Nuclear Engineering and Design. 2020; 365. DOI: 10.1016/j.nucengdes.2020.110718
2. Lee J, Tolman M, Borrelli RA. High reliability safeguards approach to remotely handled nuclear processing facilities: Use of discrete event simulation for material throughput for fuel fabrication. Nuclear Engineering and Design. 2017; 324:54.
3. Tacke J, Borrelli R, Roberson D. Advanced frequency-domain compensator design for subsystems within a nuclear generating station. Progress in Nuclear Energy. 2021; 140. DOI: 10.1016/j.pnucene.2021.103914
4. Root SJ, Throckmorton P, Tacke J, Benjamin J, Haney M, Borrelli RA. Cyber Hardening of Nuclear Power Plants with Real-time Nuclear Reactor Operation – 1. Preliminary Operational Testing. Progress in Nuclear Energy. 2023; 162:104742.
5. Borrelli R, Araujo K, Koerner C, Djokic D. Consent based siting for Spent Nuclear Fuel – The Common Ground Consortium Focus on Research and Public Conversations. Proc., American Nuclear Society Annual Meeting. 2024.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Borrelli, R. A. in SciENcv on 2026-05-07 16:17:47